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Digital Humanities

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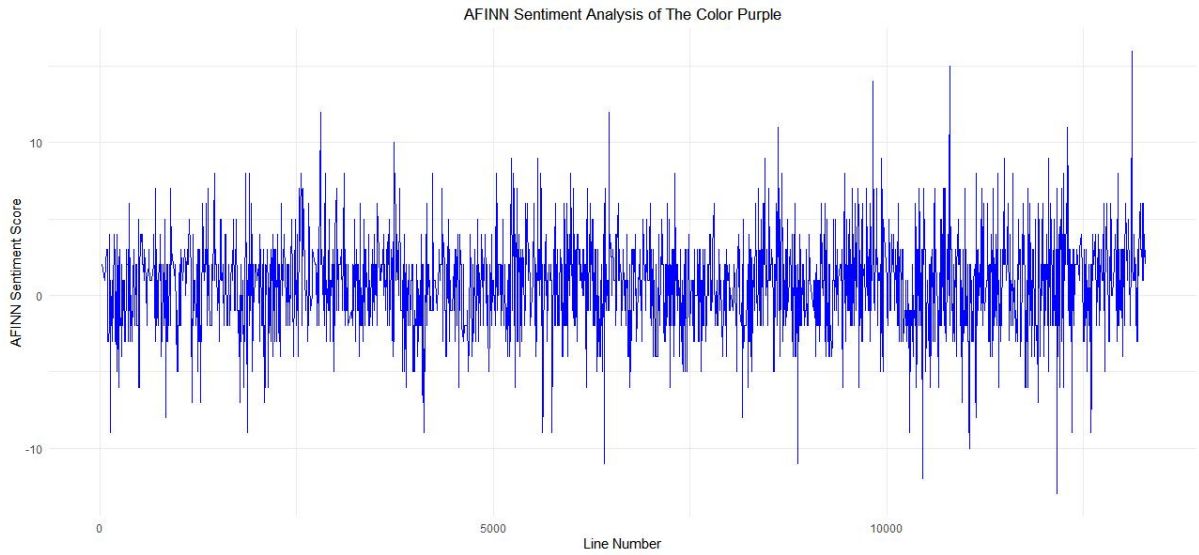
Emotion analysis of *The Color Purpl*e using R

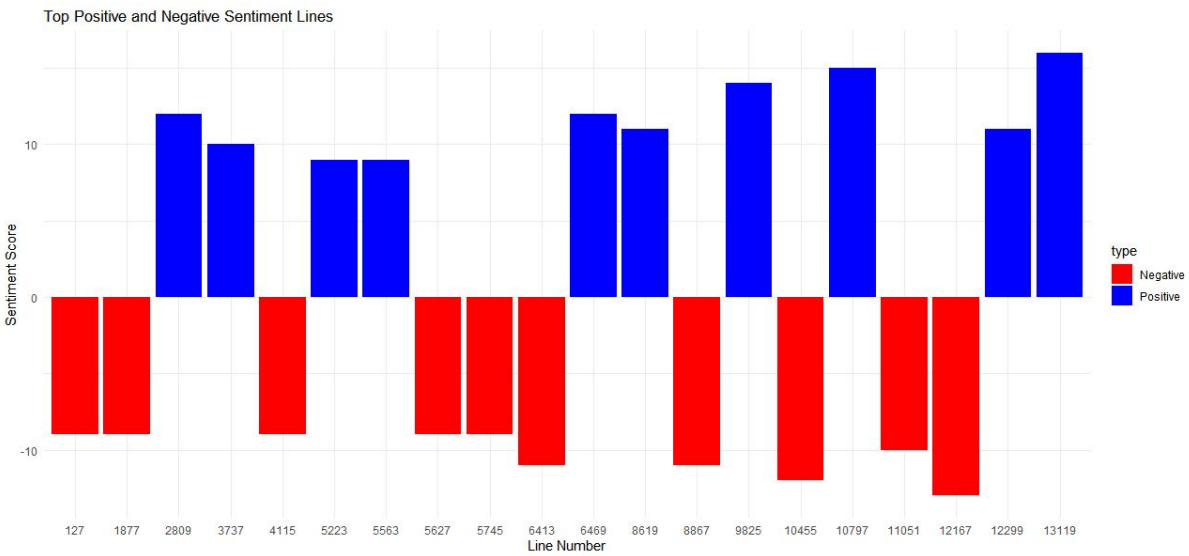
The Color Purple is an important work by Alice Walker that delves into the experiences of a young Black woman in the early 20th century American South. The story is told through a collection of letters, showcasing the growth of the main character, Celie, as she navigates issues related to race, gender, family, and her own identity. To better understand the sentiment depth of the novel, we utilize different emotion dictionaries to evaluate the sentiment scores and perform a visual analysis of these feelings.

I. AFINN Sentiment Analysis of *The Color Purple*

Firstly, I will conduct an analysis of the text from *The Color Purple* utilizing AFFIN's sentiment dictionary.



1.1 AFINN Dictionary

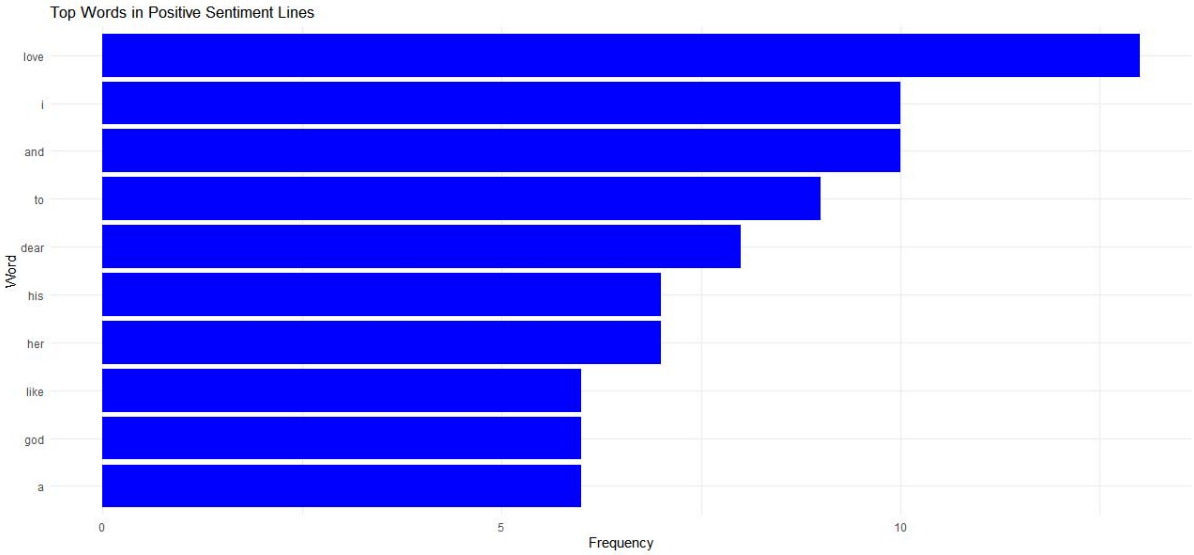


1.2 AFINN Dictionary of Top Positive and Negatve Sentiment Lines

The sentiment dynamics exhibited by *The Color Purple* encompass a broad spectrum of fluctuations, ranging from -10 to +10, characterized by a pronounced alternation between positive and negative emotions. This variability mirrors the dramatic narrative arc of the novel. The high frequency of these sentiment shifts suggests a tightly woven storyline. Notably, around the 6000-line mark, the sentiment apex aligns with a pivotal moment in the evolution of Celie and Shug's relationship, where the intensity of sentiment fluctuations escalates, thereby illustrating the complexity inherent in their character interactions. This segment indicates a trajectory toward sentiment peaks, signifying Celie's nascent personal development.

From the standpoint of sentiment progression, the initial phase is marked by a predominance of negative emotions, which reflects Celie's state of despair. In the subsequent phase, a rise in positive sentiment peaks emerges, indicating a turning point in the characters' destinies. The frequency of sentiment oscillations increases in the latter part of the narrative, underscoring the richness of the plot. In terms of narrative structure analysis, the intensity of sentiment fluctuations serves as an indicator of the narrative rhythm, with significant plot developments frequently coinciding with notable sentiment shifts. Furthermore, the evolution

of character relationships contributes to an overarching transformation in the sentiment tone, enhancing the overall engagement of the narrative.



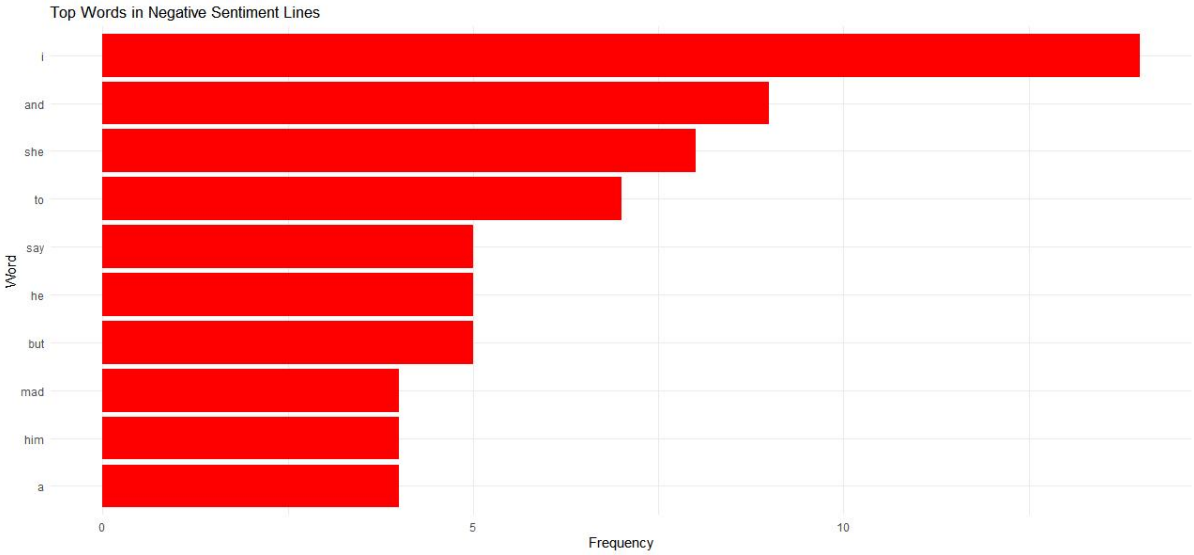
1.3 Positive word frequency

The analysis of word frequency in positive sentiment texts from *The Color Purple* indicates that the term "love" is the most prevalent, signifying its role as a central theme within the narrative. This focus on love accentuates the profound sentiment connections and relationships among the characters. Additionally, the frequent occurrence of words such as "I," "and," "to," and "her" further emphasizes personal expressions of emotion and the significance of interpersonal relationships, suggesting intimate dialogues and interactions among the characters.

Moreover, the inclusion of terms like "dear" and "like" conveys a sense of closeness and appreciation, reflecting the valued bonds between characters. The mention of "god" introduces a spiritual dimension, alluding to themes of faith and resilience that are frequently examined in literary works.

In summary, the aggregation of these high-frequency words contributes to a positive sentiment atmosphere in *The Color Purple*, highlighting the power of love, the importance of human connections, and the enduring hope that exists even in the face of adversity. This

analysis not only illuminates the characters' challenges but also emphasizes the restorative potential of love and relationships.



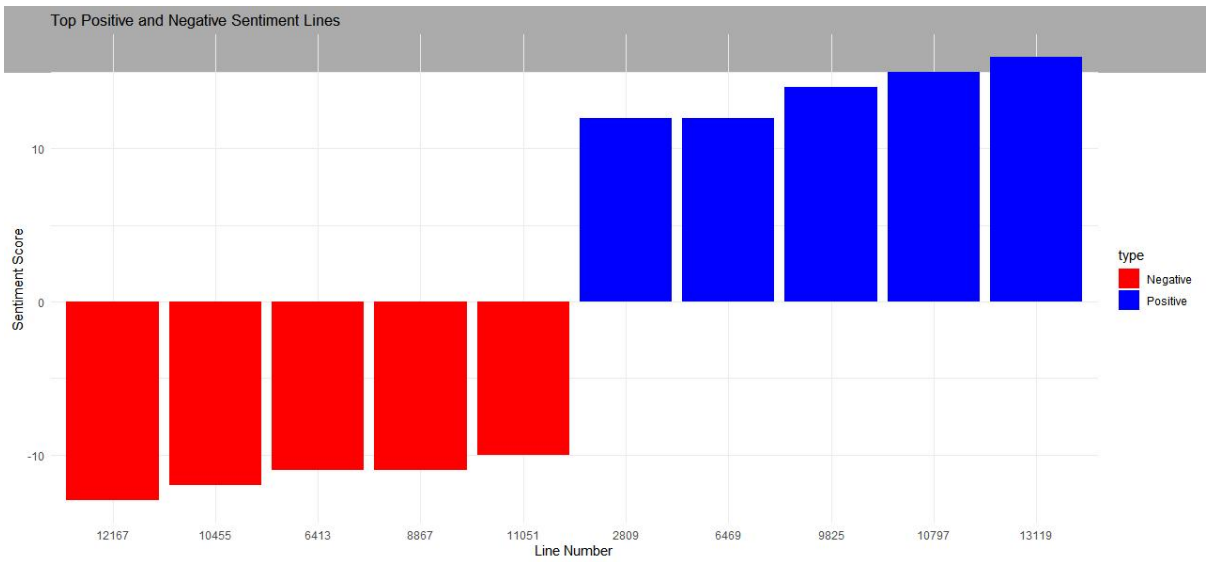
1.4 Negative word frequency

An analysis of word frequency pertaining to negative sentiment content reveals that demonstrative and personal pronouns, particularly "I," are the most frequently utilized terms in *The Color Purple*. This prevalence suggests a significant emphasis on individual experiences and personal adversities within the narrative. The dominance of first-person pronouns indicates that characters frequently articulate their thoughts and emotions directly, which may reflect a sense of isolation or introspection.

Additionally, other high-frequency terms such as "and," "to," and "she" further underscore the relational dynamics present in the text, suggesting interactions that may possess a negative or confrontational undertone. The presence of words like "but," "mad," and "him" serves to highlight themes of conflict and sentiment distress, indicating discord in interpersonal relationships and internal struggles faced by the characters.

In summary, the frequency of these terms within lines characterized by negative sentiment illustrates a landscape of struggle, frustration, and sentiment anguish in *The Color Purple*. This examination emphasizes the characters' challenges and the more somber

dimensions of their experiences, which stand in stark contrast to the themes of love and connection evident in the positive sentiment text. The emphasis on personal pronouns reinforces the notion that the narrative profoundly investigates individual suffering and the intricacies of human relationships.



1.5 Top Positive and Negatve Sentiment Lines

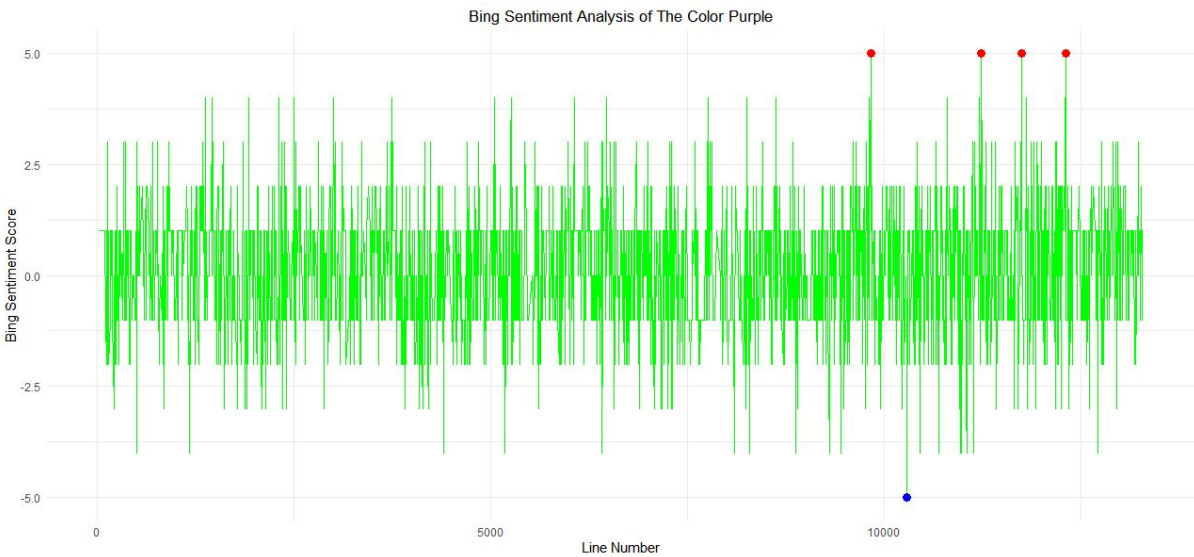
The examination of sentiment distribution within *The Color Purple* reveals a significant trend: a predominance of positive sentiment lines prior to line 8000, contrasted with an increase in negative sentiment lines in the latter portion of the text. This alteration in sentiment tone signifies a narrative transition, wherein the initial segments predominantly highlight themes of love, hope, and connection, thereby reflecting the characters' relationships and favorable experiences.

As the narrative unfolds, the heightened occurrence of negative sentiment lines indicates that the characters encounter more substantial challenges, conflicts, and sentiment turmoil. This juxtaposition between the earlier positive passages and the subsequent negative ones underscores the intricacies of the characters' journeys, illustrating their navigation through both uplifting and distressing experiences.

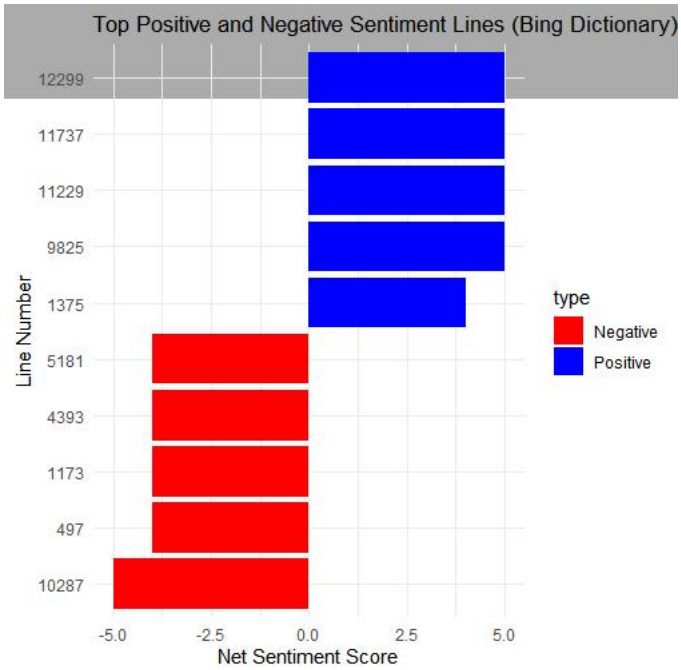
To sum up, this sentiment distribution underscores the dynamic nature of the narrative, illustrating the transformation of the characters' experiences over time and the complexity of their sentiment trajectories. The transition from positivity to negativity amplifies the significance of their struggles, thereby rendering the themes of resilience and personal growth even more impactful.

II. Bing Sentiment Analysis of *The Color Purple*

Secondly, I used the BING Sentiment Dictionary to conduct sentiment analysis on *The Color Purple.*



2.1 Bing Sentiment Analysis



2.2 Bing Dictionary

The visual analysis chart derived from the Bing Emotion Dictionary reveals the fluctuations in sentiment present in *The Color Purple*. The emotion scores range from -6 to +5, indicating a broadly varied emotional landscape. The graph illustrates significant variations in emotional intensity, with scores frequently oscillating between negative and positive values. This pattern highlights the rich emotional transitions depicted in the narrative, reflecting the complex emotional experiences of the characters. The frequency and magnitude of these sentiment fluctuations further suggest a tightly woven narrative rhythm, particularly evident during pivotal plot developments, where sentiment scores exhibit more pronounced shifts, thereby enhancing the dramatic quality of the story.

In the analysis, red dots signify segments of text with elevated sentiment scores, while blue dots denote segments with diminished sentiment scores, representing the strong emotions expressed within the novel. Specific passages identified as exhibiting high emotional intensity include lines 9825, 11229, 11737, and 12999, which correspond to the following excerpts:

1. “Suzie Q so proud of her, she say. Love her singing. Love .her perfume. Love her dresses. Love to wear her hats and shoes.”

2. “255 cookies. They so cute. And I read my fortune right away. It say, because you are who you are, the future look happy and bright.”

3. “to love. No honey colored curly hair, no cuteness. Nothing young and fresh. My heart must be young and fresh though, it feel like”

4. “been love. And I thank God he let me gain”

5. “bright, but she ain't that bright. Gear medium brown skin, gleam on it like on good furniture”

Conversely, the lines associated with strong low emotional sentiment are 10,287; 497; 1,173; 4,493; and 5,181, with the corresponding excerpts as follows:

1. “And as they struggled to put up roofs of this cold, hard, glittery, ugly metal the women raised a deafening ululation of sorrow”

2. “hair. They only six and eight and they cry. They scream. They cuse me of murder. By ten o'clock I'm done. They cry themselves”

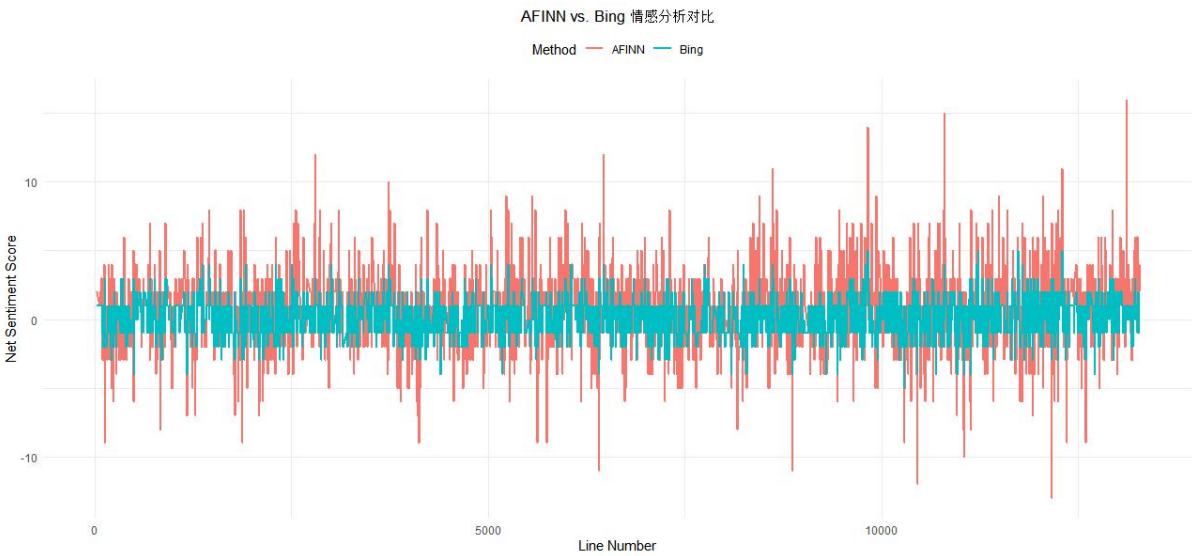
3. “throw himself on the bed. He tired. He sad. He weak. He cry. Then he sleep the rest of the day and all night.”

4. “Poor little Squeak come home with a limp. Her dress rip. Her hat missing and one of the heels come off her shoe.”

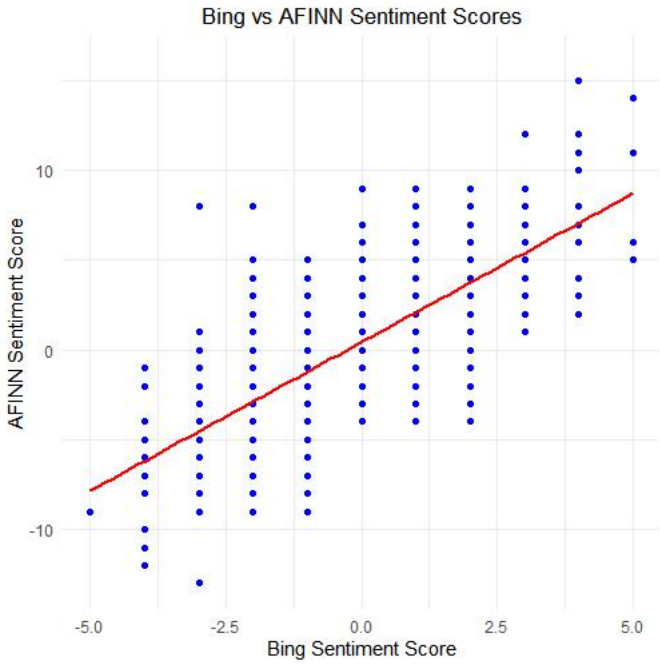
5. “I start to cry too. I cry and cry and cry. Seem like it all come back to me, laying there in Shug arms. How it hurt and how much”

After that, word frequency statistics were performed on high-sentiment texts and low-sentiment texts, and the results were as follows. The results were not significant and thus no detailed analysis was performed.

III. AFINN vs.Bing Sentiment Analysis Comparison



3.1 AFINN vs.Bing Sentiment analysis comparison



3.2 Bing vs AFINN Sentiment Scores

The sentiment analysis of *The Color Purple* utilizing both the AFINN and Bing methodologies yields significant insights into the emotional dimensions of the text. The correlation coefficient of 0.767 between the two approaches indicates a robust consistency in their sentiment assessments. Both methodologies effectively capture the emotional fluctuations inherent in the narrative, notably emphasizing a marked increase in sentiment during Celie's interactions with Shug in the latter portion of the story.

Graph 3.1 illustrates the positive correlation between the sentiment scores derived from Bing and AFINN, thereby affirming that both methods produce analogous trends in emotional evaluation. Graph 3.2 offers a comprehensive depiction of sentiment scores across various lines of text, highlighting the emotional variations and underscoring the complexity of the characters' feelings.

In conclusion, irrespective of the analytical method employed, the sentiment analysis reveals considerable emotional variation within *The Color Purple*, particularly accentuating the depth and intricacies of the characters' emotions as the narrative unfolds.

IV. Conclusion

In conclusion, the sentiment analysis of *The Color Purple*, both AFINN and Bing methodologies offers a comprehensive insight into the emotional dimensions of the novel. The AFINN analysis reveals a broad spectrum of sentiment variations that correspond with the dramatic narrative trajectory and character development, particularly in the relationship between Celie and Shug. The initial prevalence of negative sentiments gradually transitions to positive emotions, effectively illustrating Celie's journey toward self-realization and empowerment.

In contrast, the Bing analysis supports these observations by revealing significant emotional fluctuations that heighten the narrative's dramatic intensity. The correlation coefficient of 0.767 between the two analytical approaches highlights their alignment in capturing the emotional dynamics, especially during crucial plot developments.

Overall, these analyses illuminate the intricate interplay of love, struggle, and personal growth throughout the narrative of *The Color Purple*. The varying sentiment distributions highlight the characters' resilience, reflecting their experiences of both uplifting and challenging moments. This nuanced examination of emotions not only deepens our

understanding of the characters' journeys but also reinforces the enduring themes of love and connection in the face of adversity within Alice Walker's powerful narrative.